Chapter 3 • Sets and Functions

Solved Exercise 3.3

1. For A = {1,2,3,4}, list each relation in A and state its domain and range.

(i) y=x: {(1,1),(2,2),(3,3),(4,4)}
domain = range
= {1,2,3,4}.
(ii) x+y=5: {(1,4),(2,3),(3,2),(4,1)}
domain = range
= {1,2,3,4}.
(iii) x+y<5: {(1,1),(1,2),(1,3),(2,1),(2,2),(3,1)}
domain = range
= {1,2,3}.
(iv) x+y>5: {(2,4),(3,3),(3,4),(4,2),(4,3),(4,4)}
domain = range
= {2,3,4}.

2. Which arrow diagrams represent functions, and of which type?

Fig (1): not a function because input 1 has more than one image.
Fig (2): a function; a→1, b→3, c→5. It is one-to-one but not onto because 2 and 4 are unhit.
Therefore it is injective and into, not bijective.
Fig (3): a function; 1→a, 2→b, 3→c. It is one-to-one and onto, hence bijective.
Fig (4): a function; l→x, m→x, n→z. It is many-to-one and into because y is unhit.
(1) not a function; (2) one-to-one into; (3) bijective; (4) many-to-one into.

3. For g(x)=3x+2 and h(x)=x²+1, evaluate the six expressions.

(i) g(0)=2
(ii) g(−3)=−7
(iii) g(2/3)=4
(iv) h(1)=2
(v) h(−4)=17
(vi) h(−1/2) = $\frac{1}{4}+1$
= $\frac{5}{4}$

4. f(x)=ax+b+1, f(3)=8 and f(6)=14. Find a and b.

3a+b+1=8
⇒ 3a+b=7
6a+b+1=14
⇒ 6a+b=13.
Subtracting gives 3a=6 ⇒ a=2; then b=7−6=1.
a=2, b=1.

5. g(x)=ax+b+5, g(−1)=0 and g(2)=10. Find a and b.

−a+b+5=0
⇒ −a+b=−5
2a+b+5=10
⇒ 2a+b=5.
Subtracting gives 3a=10 ⇒ a=$\frac{10}{3}$; then b=a−5=$\frac{-5}{3}$
a=$\frac{10}{3}$, b=$\frac{-5}{3}$

6. For f(x)=5x+2, find x when f(x)=32.

5x+2=32
⇒ 5x=30
⇒ x=6.

7. f(x)=cx²+d, f(1)=6 and f(−2)=10. Find c and d.

c+d=6 and 4c+d=10. Subtracting gives 3c=4.
c=$\frac{4}{3}$
d = 6−$\frac{4}{3}$
= $\frac{14}{3}$

studysapiens.com

$c=\frac{4}{3}$, $d=\frac{14}{3}$

studysapiens.com